

I Wanna Dance With Somebody: Generating Authentic Crowds With Volumetric Video

Callum Macmillan
Dimension
London, UK
callum@dimensionstudio.co

Don Libby
Zero VFX
Boston, Massachusetts, USA
dlibby@zerovfx.com

Brian Drewes
Zero VFX
Boston, Massachusetts, USA
brian@zerovfx.com

Eric Robertson
ReDefine
Los Angeles, California, USA
erob@redefine.co



Figure 1: (left) Volumetric video shoot and volumetric video. (right) Crowd scene from the film “I Wanna Dance With Somebody. All Rights Reserved.

ABSTRACT

At the center of Whitney Houston biopic “I Wanna Dance With Somebody” are performances with crowds upwards of 70,000 people, taking place over a period of 40 years. This talk explores a collaborative new process using volumetric video capture for generating crowds for feature film, with each extra captured in 3D, allowing for authentic movement and emotion, expansive camera moves and avoiding the use of thousands of extras.

ACM Reference Format:

Callum Macmillan, Brian Drewes, Don Libby, and Eric Robertson. 2023. I Wanna Dance With Somebody: Generating Authentic Crowds With Volumetric Video. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks (SIGGRAPH '23 Talks)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3587421.3595461>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
SIGGRAPH '23 Talks, August 06–10, 2023, Los Angeles, CA, USA
© 2023 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0143-6/23/08.
<https://doi.org/10.1145/3587421.3595461>

1 INTRODUCTION

Creating authentic crowds for feature film scenes has historically involved using extra performers, in numbers up to 69,000 (Around the World in 80 Days, 1956). In recent years, digital techniques have simulated large crowds, with limited results. Zero and Dimension have collaborated to create a new pipeline that uses volumetric video capture to generate crowds. For “I Wanna Dance With Somebody,” performers were each captured multiple times in 2K and 4K, resulting in 510 minutes of volumetric footage and a total of 1,000 captures. The new pipeline allowed for motion blur and prop replacement. This 3D technique meant stadiums could be filled with over 70,000 people, with authentic movement of clothing and facial expressions. The data was then shared with ReDefine to digitally create large crowds in massive concert venues including Wembley Stadium, Palais Omnisports de Paris-Bercy and Arena Nürnberger Versicherung.

2 PREPARATION AND ACQUISITION

Volumetric video captures human performance from all angles. The result is a 3D video, allowing the user to view any point of the performance from any angle. General applications of volumetric video are for VFX & ICVFX pipelines, augmented reality, virtual reality and mixed reality. With a number of previous productions having unsuccessfully explored the use of volumetric video for crowds, pipelines and methodology were validated and stress-tested for a

year ahead of principal photography in preparation for “I Wanna Dance With Somebody.” Producers studied existing footage of real performances, recording emotions, costumes and movements.

2.1 Volumetric video and new capture tools

A 70-camera volumetric capture stage was built in Boston, including 39 RGB cameras which read and record the colour required for the .png texture map and 31 infra-red cameras which record depth and position in space for creating the mesh. Video was captured in both 2K and 4K. Over a two-week period, 300 actors were each captured volumetrically in 3D several times, resulting in a total of 1,000 performances. This was the world’s largest volumetric video shoot to date. In-volume prop tracking enabled digital prop replacement in post without matchmoving, while a motion vector solution was created to ensure realistic motion blur, a key discovery required for realistic integration into scenes. The performers were globally lit, enabling the VFX team to relight them according to the scene.

3 PROCESSING & EDITORIAL

Using Microsoft Mixed Reality Capture Solver, a tech art team processed 510 minutes of volumetric video footage (700 terabytes of data) over an eight-week period, before delivering 30 terabytes of processed assets to Zero and then to ReDefine. Each volumetric capture was then checked by Quality Control, before being taken into a Houdini driven pipeline. Each capture was tagged to allow Houdini to disperse crowd members, with tags corresponding to actions, character ID, and the concert location they were captured for. The pipeline was optimized to avoid collisions and allowed switching out of individual audience members where required. Shading and compositing pipelines were established to allow for further variations of the colors of costumes and skin tones. For example, if there were too many shirts of the same color in an area, the color of each asset’s clothing could be isolated and shifted in the composite, without the need to shift the arrangement of crowd members and re-render the CG. The ability to do this became pivotal in ensuring that precious render resources were being effectively allocated to where they were needed most. To make sure these volumetric crowd assets were properly placed into their scenes, each major concert venue was LiDAR scanned in an extremely high fidelity, which both made the 3D tracking of each shot much more accurate as well as gave a great base for the expanded stadium builds that were needed to match the real-life filming locations to the locations historically accurate to Whitney’s various performances featured in the film.

4 POST-PRODUCTION

With all the processing and organization handled ahead of time, once the editorial team was ready to turn over shots it was a very seamless system to fully populate the crowd after 3D tracking had been completed. Using a combination of HDRIs captured on set and hand placed lighting, the volumetric crowds could quickly be lit to match each individual shot, owing to how evenly lit the extras were when captured.

Because of the natural nuance that comes from capturing real people instead of using an animation driven crowd system, the shots were very realistic and believable from an early stage. From

far distances you had a crowd in sync with the actions necessary for the emotion of the scene, while not feeling robotic or repetitive. Close up you had the subtle actions of clothing wrinkling in motion, micro facial expressions, and small details that wouldn’t be possible without complex facial rigs, motion capture, and heavy to manage cloth simulations. The shots were far from done straight out of the gate, but with the crowds being so realistic from the start it allowed the creative team to worry more about the story they were trying to tell rather than being concerned about whether the technology of a more traditional crowd system would convincingly hold up in their most challenging shots.

The pipeline established allowed for a lot of flexibility in addressing creative notes from the director and editorial team. If we needed a specific action for the crowd, our Houdini tagging system could be quickly reshuffled to use the assets doing those specified actions. If we didn’t like an individual performance in the crowd that one person could be swapped out for another, or if necessary for continuity that person could be swapped for the same character performing a more muted action.

Due to the peace of mind afforded by the volumetric crowd being so natural and easily managed, the director and creative team didn’t shy away from using VFX to expand their vision. In some cases, this meant placing Whitney in a shot with a wider field of view of the crowd, to make the performance feel grander and more spectacular like the real life crowds Whitney was able to draw in. In other instances, full CG shots were designed to create moments that weren’t possible to capture on set; a blimp shot of the Super Bowl stadium, the camera pulling back and sweeping over the heads of the crowd, and fireworks bursting in time with one of Whitney’s most iconic songs.